

Probability & Random Processes EE2011

Department of Electrical Engineering

**Joint Probability Distribution, Marginal Distribution, Conditional Distribution,
Statistical Independence**

NATIONAL UNIVERSITY OF COMPUTER & EMERGING
SCIENCES, LAHORE CAMPUS



Definition 3.1:

A **random variable** is a function that associates a real number with each element in the sample space.

Definition 3.2:

If a sample space contains a finite number of possibilities or an unending sequence with as many elements as there are whole numbers, it is called a **discrete sample space**.

Definition 3.3:

If a sample space contains an infinite number of possibilities equal to the number of points on a line segment, it is called a **continuous sample space**.



Definition 3.4:

The set of ordered pairs $(x, f(x))$ is a **probability function**, **probability mass function**, or **probability distribution** of the discrete random variable X if, for each possible outcome x ,

1. $f(x) \geq 0$,
2. $\sum_x f(x) = 1$,
3. $P(X = x) = f(x)$.

Definition 3.5:

The **cumulative distribution function** $F(x)$ of a discrete random variable X with probability distribution $f(x)$ is

$$F(x) = P(X \leq x) = \sum_{t \leq x} f(t), \quad \text{for } -\infty < x < \infty.$$



Definition 3.6:

The function $f(x)$ is a **probability density function** (pdf) for the continuous random variable X , defined over the set of real numbers, if

1. $f(x) \geq 0$, for all $x \in R$.
2. $\int_{-\infty}^{\infty} f(x) dx = 1$.
3. $P(a < X < b) = \int_a^b f(x) dx$.



Definition 3.7:

The **cumulative distribution function** $F(x)$ of a continuous random variable X with density function $f(x)$ is

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt, \quad \text{for } -\infty < x < \infty.$$

As an immediate consequence of Definition 3.7, one can write the two results

$$P(a < X < b) = F(b) - F(a) \text{ and } f(x) = \frac{dF(x)}{dx},$$

if the derivative exists.



Example 1

Classify the following random variables as discrete or continuous:

X : the number of automobile accidents per year in Virginia.

Y : the length of time to play 18 holes of golf.

M : the amount of milk produced yearly by a particular cow.

N : the number of eggs laid each month by a hen.

P : the number of building permits issued each month in a certain city.

Q : the weight of grain produced per acre.



Example 2

3.12 An investment firm offers its customers municipal bonds that mature after varying numbers of years. Given that the cumulative distribution function of T , the number of years to maturity for a randomly selected bond, is

$$F(t) = \begin{cases} 0, & t < 1, \\ \frac{1}{4}, & 1 \leq t < 3, \\ \frac{1}{2}, & 3 \leq t < 5, \\ \frac{3}{4}, & 5 \leq t < 7, \\ 1, & t \geq 7, \end{cases}$$

find

- (a) $P(T = 5)$;
- (b) $P(T > 3)$;
- (c) $P(1.4 < T < 6)$;
- (d) $P(T \leq 5 \mid T \geq 2)$.



Example 2

Solution Example 2:

(a) $P(T = 5) = F(5) - F(4) = 3/4 - 1/2 = 1/4.$

(b) $P(T > 3) = 1 - F(3) = 1 - 1/2 = 1/2.$

(c) $P(1.4 < T < 6) = F(6) - F(1.4) = 3/4 - 1/4 = 1/2.$



Example 3

3.13 The probability distribution of X , the number of imperfections per 10 meters of a synthetic fabric in continuous rolls of uniform width, is given by

x	0	1	2	3	4
$f(x)$	0.41	0.37	0.16	0.05	0.01

Construct the cumulative distribution function of X .

Solution Example 3:

The c.d.f. of X is

$$F(x) = \begin{cases} 0, & \text{for } x < 0, \\ 0.41, & \text{for } 0 \leq x < 1, \\ 0.78, & \text{for } 1 \leq x < 2, \\ 0.94, & \text{for } 2 \leq x < 3, \\ 0.99, & \text{for } 3 \leq x < 4, \\ 1, & \text{for } x \geq 4. \end{cases}$$



Example 4

3.17 A continuous random variable X that can assume values between $x = 1$ and $x = 3$ has a density function given by $f(x) = 1/2$.

- (a) Show that the area under the curve is equal to 1.
- (b) Find $P(2 < X < 2.5)$.
- (c) Find $P(X \leq 1.6)$.

Solution Example 4

(a) $\text{Area} = \int_1^3 (1/2) \, dx = \left. \frac{x}{2} \right|_1^3 = 1.$

(b) $P(2 < X < 2.5) = \int_2^{2.5} (1/2) \, dx = \left. \frac{x}{2} \right|_2^{2.5} = \frac{1}{4}.$

(c) $P(X \leq 1.6) = \int_1^{1.6} (1/2) \, dx = \left. \frac{x}{2} \right|_1^{1.6} = 0.3.$



3.4 Joint Probability Distributions

There are situations, where we find it desirable to record the simultaneous outcomes of several random variables.

Definition 3.8:

The function $f(x, y)$ is a **joint probability distribution** or **probability mass function** of the discrete random variables X and Y if

1. $f(x, y) \geq 0$ for all (x, y) ,
2. $\sum_x \sum_y f(x, y) = 1$,
3. $P(X = x, Y = y) = f(x, y)$.

For any region A in the xy plane, $P[(X, Y) \in A] = \sum_A f(x, y)$.

When X and Y are continuous random variables, the **joint density function** $f(x, y)$ is a surface lying above the xy plane, and $P[(X, Y) \in A]$, where A is any region in the xy plane, is equal to the volume of the right cylinder bounded by the base A and the surface.



Definition 3.9:

The function $f(x, y)$ is a **joint density function** of the continuous random variables X and Y if

1. $f(x, y) \geq 0$, for all (x, y) ,
2. $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) \, dx \, dy = 1$,
3. $P[(X, Y) \in A] = \int \int_A f(x, y) \, dx \, dy$, for any region A in the xy plane.



EXAMPLE 5: Discrete RV Joint PMF

Consider two random variables X and Y with joint PMF given in Table 5.1.

Table 5.1 Joint PMF of X and Y in Example 5.1

	$Y = 0$	$Y = 1$	$Y = 2$
$X = 0$	$\frac{1}{6}$	$\frac{1}{4}$	$\frac{1}{8}$
$X = 1$	$\frac{1}{8}$	$\frac{1}{6}$	$\frac{1}{6}$

- Find $P(X = 0, Y \leq 1)$.
- Find the marginal PMFs of X and Y .
- Find $P(Y = 1|X = 0)$.
- Are X and Y independent?

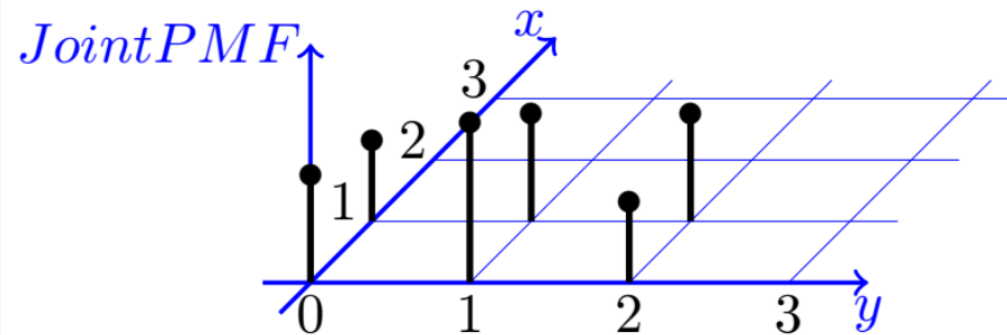
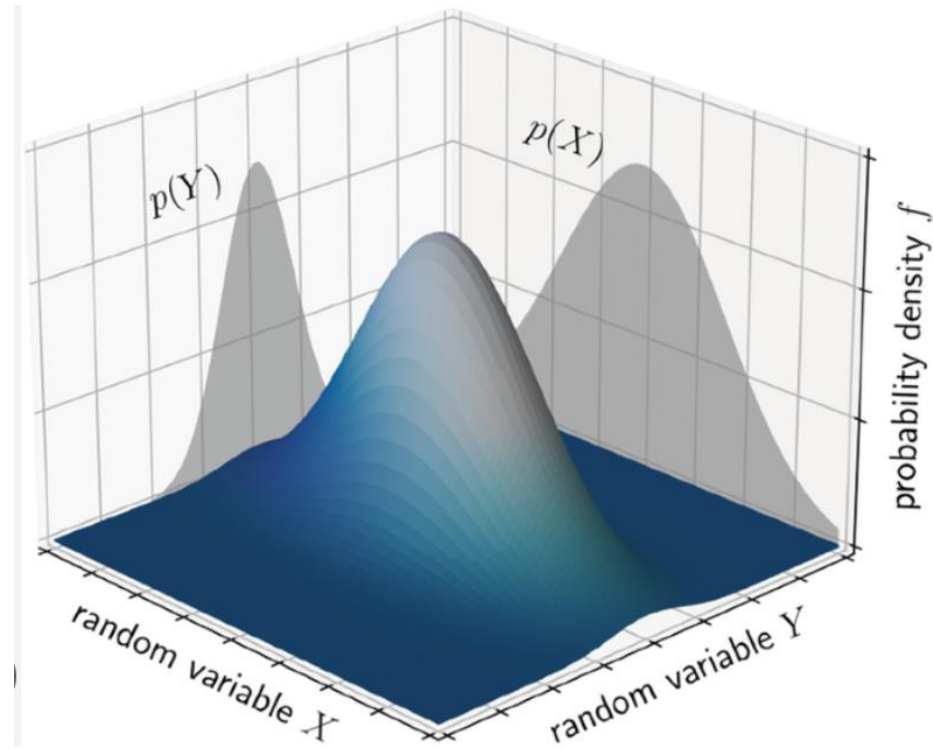


Figure 5.1: Joint PMF of X and Y (Example 5.1).





Example 3.14:

Two ballpoint pens are selected at random from a box that contains 3 blue pens, 2 red pens, and 3 green pens.

If X is the number of blue pens selected and

Y is the number of red pens selected,

find

(a) the joint probability function $f(x, y)$,

(b) $P[(X, Y) \in A]$, where A is the region $\{(x, y) | x + y \leq 1\}$.

Solution:

The possible pairs of values (x, y) are $(0, 0)$, $(0, 1)$, $(1, 0)$, $(1, 1)$, $(0, 2)$, and $(2, 0)$.



- (a) Now, $f(0, 1)$, for example, represents the probability that a red and a green pens are selected.

The total number of equally likely ways of selecting any 2 pens from the 8 is $\binom{8}{2} = 28$.

The number of ways of selecting 1 red from 2 red pens and 1 green from 3 green pens is $\binom{2}{1} \binom{3}{1} = 6$.

Hence, $f(0, 1) = 6/28 = 3/14$.

Similar calculations yield the probabilities for the other cases, which are presented in Table 3.1.



Table 3.1: Joint Probability Distribution for Example 3.14

		x		
		0	1	2
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0
	2	$\frac{1}{28}$	0	0

Note that the probabilities sum to 1.

the joint probability distribution of Table 3.1 can be represented by the formula

$$f(x, y) = \frac{\binom{3}{x} \binom{2}{y} \binom{3}{2-x-y}}{\binom{8}{2}},$$

for $x = 0, 1, 2$; $y = 0, 1, 2$; and $0 \leq x + y \leq 2$.



(b) The probability that (X, Y) fall in the region A is

$$P[(X, Y) \in A] = P(X + Y \leq 1) = f(0, 0) + f(0, 1) + f(1, 0)$$

$$= \frac{3}{28} + \frac{3}{14} + \frac{9}{28} = \frac{9}{14}.$$



Example 3.15:

A privately owned business operates both a drive-in facility and a walk-in facility.

On a randomly selected day, let X and Y , respectively, be the proportions of the time that the drive-in and the walk-in facilities are in use, :

and suppose that the joint density function of these random variables is

$$f(x, y) = \begin{cases} \frac{2}{5}(2x + 3y), & 0 \leq x \leq 1, 0 \leq y \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

(a) Verify condition 2 of Definition 3.9.

Solution:

(a) The integration of $f(x, y)$ over the whole region is

$$\begin{aligned} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) \, dx \, dy &= \int_0^1 \int_0^1 \frac{2}{5}(2x + 3y) \, dx \, dy \\ &= \int_0^1 \left(\frac{2x^2}{5} + \frac{6xy}{5} \right) \Big|_{x=0}^{x=1} dy \\ &= \int_0^1 \left(\frac{2}{5} + \frac{6y}{5} \right) dy = \left(\frac{2y}{5} + \frac{3y^2}{5} \right) \Big|_0^1 = \frac{2}{5} + \frac{3}{5} = 1. \end{aligned}$$



(b) Find $P[(X, Y) \in A]$, where $A = \{(x, y) \mid 0 < x < \frac{1}{2}, \frac{1}{4} < y < \frac{1}{2}\}$.

(b) To calculate the probability, we use

$$\begin{aligned} P[(X, Y) \in A] &= P\left(0 < X < \frac{1}{2}, \frac{1}{4} < Y < \frac{1}{2}\right) \\ &= \int_{1/4}^{1/2} \int_0^{1/2} \frac{2}{5}(2x + 3y) \, dx \, dy \\ &= \int_{1/4}^{1/2} \left(\frac{2x^2}{5} + \frac{6xy}{5}\right) \Big|_{x=0}^{x=1/2} dy = \int_{1/4}^{1/2} \left(\frac{1}{10} + \frac{3y}{5}\right) dy \\ &= \left(\frac{y}{10} + \frac{3y^2}{10}\right) \Big|_{1/4}^{1/2} \\ &= \frac{1}{10} \left[\left(\frac{1}{2} + \frac{3}{4}\right) - \left(\frac{1}{4} + \frac{3}{16}\right)\right] = \frac{13}{160}. \end{aligned}$$



Given the joint probability distribution $f(x, y)$ of the discrete random variables X and Y , the probability distribution $g(x)$ of X alone is obtained by summing $f(x, y)$ over the values of Y .

Similarly, the probability distribution $h(y)$ of Y alone is obtained by summing $f(x, y)$ over the values of X .

We define $g(x)$ and $h(y)$ to be the **marginal distributions** of X and Y , respectively.

When X and Y are continuous random variables, summations are replaced by integrals.

We can now make the following general definition.



Definition 3.10:

The **marginal distributions** of X alone and of Y alone are

$$g(x) = \sum_y f(x, y) \quad \text{and} \quad h(y) = \sum_x f(x, y)$$

for the discrete case, and

$$g(x) = \int_{-\infty}^{\infty} f(x, y) dy \quad \text{and} \quad h(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

for the continuous case.



Example 3.16:

Show that the column and row totals of Table 3.1 give the marginal distribution of X alone and of Y alone.

Solution:

For the random variable X , we see that

$$g(x) = \sum_y f(x, y) \quad \text{and} \quad h(y) = \sum_x f(x, y)$$

		x		
		0	1	2
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0
	2	$\frac{1}{28}$	0	0

$$g(0) = f(0, 0) + f(0, 1) + f(0, 2) = \frac{3}{28} + \frac{3}{14} + \frac{1}{28} = \frac{5}{14},$$

$$g(1) = f(1, 0) + f(1, 1) + f(1, 2) = \frac{9}{28} + \frac{3}{14} + 0 = \frac{15}{28},$$

and $g(2) = f(2, 0) + f(2, 1) + f(2, 2) = \frac{3}{28} + 0 + 0 = \frac{3}{28},$

which are just the column totals of Table 3.1.

x	0	1	2
$g(x)$	$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$

In a similar manner we could show that the values of $h(y)$ are given by the row totals.

y	0	1	2
$h(y)$	$\frac{15}{28}$	$\frac{3}{7}$	$\frac{1}{28}$



$f(x, y)$		x			Row Totals
		0	1	2	
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$	$\frac{15}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0	$\frac{3}{7}$
	2	$\frac{1}{28}$	0	0	$\frac{1}{28}$
Column Totals		$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$	1

The term *marginal* is used here because, in the discrete case, the values of $g(x)$ and $h(y)$ are just the marginal totals of the respective columns and rows when the values of $f(x, y)$ are displayed in a rectangular table.



$f(x, y)$		x			Row Totals
		0	1	2	
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$	$\frac{15}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0	$\frac{3}{7}$
	2	$\frac{1}{28}$	0	0	$\frac{1}{28}$
Column Totals		$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$	1

	x		
	0	1	2
$g(x)$ of X	$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$

		$h(y)$ of Y
y	0	$\frac{15}{28}$
	1	$\frac{3}{7}$
	2	$\frac{1}{28}$



Example 3.17:

Find $g(x)$ and $h(y)$ for the joint density function of Example 3.15.

$$g(x) = \int_{-\infty}^{\infty} f(x, y) dy \quad \text{and} \quad h(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

Solution:

By definition,

$$g(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_0^1 \frac{2}{5}(2x + 3y) dy = \left(\frac{4xy}{5} + \frac{6y^2}{10} \right) \Big|_{y=0}^{y=1} = \frac{4x + 3}{5},$$

for $0 \leq x \leq 1$, and $g(x) = 0$ elsewhere. Similarly,

$$h(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_0^1 \frac{2}{5}(2x + 3y) dx = \frac{2(1 + 3y)}{5},$$

for $0 \leq y \leq 1$, and $h(y) = 0$ elsewhere. └



The fact that the marginal distributions $g(x)$ and $h(y)$ are indeed the probability distributions of the individual variables X and Y alone can be verified by showing that the conditions of Definition 3.4 or Definition 3.6 are satisfied.

For example, in the continuous case

$$\int_{-\infty}^{\infty} g(x) dx = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dy dx = 1,$$

and

$$\begin{aligned} P(a < X < b) &= P(a < X < b, -\infty < Y < \infty) \\ &= \int_a^b \int_{-\infty}^{\infty} f(x, y) dy dx = \int_a^b g(x) dx. \end{aligned}$$



In Section 3.1, we stated that the value x of the random variable X represents an event that is a subset of the sample space.

If we use the definition of conditional probability as stated in Chapter 2,

$$P(B|A) = \frac{P(A \cap B)}{P(A)}, \text{ provided } P(A) > 0,$$

where A and B are now the events defined by $X = x$ and $Y = y$, respectively, then

$$P(Y = y | X = x) = \frac{P(X = x, Y = y)}{P(X = x)} = \frac{f(x, y)}{g(x)}, \text{ provided } g(x) > 0,$$

where X and Y are discrete random variables.



It is not difficult to show that the function $f(x, y)/g(x)$, which is strictly a function of y with x fixed, satisfies all the conditions of a probability distribution.

This is also true when $f(x, y)$ and $g(x)$ are the joint density and marginal distribution, respectively, of continuous random variables.

As a result, it is extremely important that we make use of the special type of distribution of the form $f(x, y)/g(x)$ in order to be able to effectively compute conditional probabilities.

This type of distribution is called a **conditional probability distribution**; the formal definition follows.



Definition 3.11:

Let X and Y be two random variables, discrete or continuous. The **conditional distribution** of the random variable Y given that $X = x$ is

$$f(y|x) = \frac{f(x, y)}{g(x)}, \text{ provided } g(x) > 0.$$

Similarly, the conditional distribution of X given that $Y = y$ is

$$f(x|y) = \frac{f(x, y)}{h(y)}, \text{ provided } h(y) > 0.$$



If we wish to find the probability that the discrete random variable X falls between a and b when it is known that the discrete variable $Y = y$,

we evaluate

$$P(a < X < b \mid Y = y) = \sum_{a < x < b} f(x|y),$$

where the summation extends over all values of X between a and b .

When X and Y are continuous, we evaluate

$$P(a < X < b \mid Y = y) = \int_a^b f(x|y) \, dx.$$



Example 3.18:

Referring to Example 3.14, find the conditional distribution of X , given that $Y = 1$, and use it to determine $P(X = 0 \mid Y = 1)$.

Solution:

We need to find $f(x|y)$, where $y = 1$.

First, we find that

$$h(1) = \sum_{x=0}^2 f(x, 1) = \frac{3}{14} + \frac{3}{14} + 0 = \frac{3}{7}.$$

Now
$$f(x|1) = \frac{f(x, 1)}{h(1)} = \left(\frac{7}{3}\right) f(x, 1), \quad x = 0, 1, 2.$$

$f(x, y)$		x			Row Totals
		0	1	2	
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$	$\frac{15}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0	$\frac{3}{7}$
	2	$\frac{1}{28}$	0	0	$\frac{1}{28}$
Column Totals		$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$	1



Therefore,

$$f(0|1) = \binom{7}{3} f(0, 1) = \binom{7}{3} \left(\frac{3}{14} \right) = \frac{1}{2},$$

$$f(1|1) = \binom{7}{3} f(1, 1) = \binom{7}{3} \left(\frac{3}{14} \right) = \frac{1}{2},$$

$$f(2|1) = \binom{7}{3} f(2, 1) = \binom{7}{3} (0) = 0,$$

and the conditional distribution of X , given that $Y = 1$, is

x	0	1	2
$f(x 1)$	$\frac{1}{2}$	$\frac{1}{2}$	0

Finally,

$$P(X = 0 | Y = 1) = f(0|1) = \frac{1}{2}.$$

		x			Row Totals
		0	1	2	
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$	$\frac{15}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0	$\frac{3}{7}$
	2	$\frac{1}{28}$	0	0	$\frac{1}{28}$
Column Totals		$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$	1



Example 3.19:

The joint density for the random variables (X, Y) , where X is the unit temperature change and Y is the proportion of spectrum shift that a certain atomic particle produces, is

$$f(x, y) = \begin{cases} 10xy^2, & 0 < x < y < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) Find the marginal densities $g(x)$, $h(y)$, and the conditional density $f(y|x)$.
- (b) Find the probability that the spectrum shifts more than half of the total observations, given that the temperature is increased by 0.25 unit.

Solution:

(a) By definition,

$$\begin{aligned} g(x) &= \int_{-\infty}^{\infty} f(x, y) dy = \int_x^1 10xy^2 dy \\ &= \frac{10}{3}xy^3 \Big|_{y=x}^{y=1} = \frac{10}{3}x(1 - x^3), \quad 0 < x < 1, \end{aligned}$$

$$h(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_0^y 10xy^2 dx = 5x^2y^2 \Big|_{x=0}^{x=y} = 5y^4, \quad 0 < y < 1.$$



Now

$$f(y|x) = \frac{f(x,y)}{g(x)} = \frac{10xy^2}{\frac{10}{3}x(1-x^3)} = \frac{3y^2}{1-x^3}, \quad 0 < x < y < 1.$$

(b) Therefore,

$$P\left(Y > \frac{1}{2} \mid X = 0.25\right) = \int_{1/2}^1 f(y \mid x = 0.25) dy = \int_{1/2}^1 \frac{3y^2}{1-0.25^3} dy = \frac{8}{9}.$$



Example 3.20:

Given the joint density function

$$f(x, y) = \begin{cases} \frac{x(1+3y^2)}{4}, & 0 < x < 2, 0 < y < 1, \\ 0, & \text{elsewhere,} \end{cases}$$

find $g(x)$, $h(y)$, $f(x|y)$, and evaluate $P(\frac{1}{4} < X < \frac{1}{2} \mid Y = \frac{1}{3})$.

Solution:

By definition of the marginal density. for $0 < x < 2$,

$$\begin{aligned} g(x) &= \int_{-\infty}^{\infty} f(x, y) dy = \int_0^1 \frac{x(1+3y^2)}{4} dy \\ &= \left(\frac{xy}{4} + \frac{xy^3}{4} \right) \Big|_{y=0}^{y=1} = \frac{x}{2}, \end{aligned}$$

and for $0 < y < 1$,

$$\begin{aligned} h(y) &= \int_{-\infty}^{\infty} f(x, y) dx = \int_0^2 \frac{x(1+3y^2)}{4} dx \\ &= \left(\frac{x^2}{8} + \frac{3x^2y^2}{8} \right) \Big|_{x=0}^{x=2} = \frac{1+3y^2}{2}. \end{aligned}$$



Therefore, using the conditional density definition, for $0 < x < 2$,

$$f(x|y) = \frac{f(x, y)}{h(y)} = \frac{x(1 + 3y^2)/4}{(1 + 3y^2)/2} = \frac{x}{2},$$

and

$$P\left(\frac{1}{4} < X < \frac{1}{2} \mid Y = \frac{1}{3}\right) = \int_{1/4}^{1/2} \frac{x}{2} dx = \frac{3}{64}.$$



Statistical Independence

If $f(x|y)$ does not depend on y , as is the case for Example 3.20, then $f(x|y) = g(x)$ and $f(x, y) = g(x)h(y)$.

The proof follows by substituting

$$f(x, y) = f(x|y)h(y)$$

into the marginal distribution of X .

That is,
$$g(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_{-\infty}^{\infty} f(x|y)h(y) dy.$$

If $f(x|y)$ does not depend on y , we may write

$$g(x) = f(x|y) \int_{-\infty}^{\infty} h(y) dy.$$

Now
$$\int_{-\infty}^{\infty} h(y) dy = 1,$$

since $h(y)$ is the probability density function of Y .

Therefore, $g(x) = f(x|y)$ and then $f(x, y) = g(x)h(y)$.



It should make sense to the reader that if $f(x|y)$ does not depend on y , then of course the outcome of the random variable Y has no impact on the outcome of the random variable X .

In other words, we say that X and Y are independent random variables.

We now offer the following formal definition of statistical independence.

Definition 3.12:

Let X and Y be two random variables, discrete or continuous, with joint probability distribution $f(x, y)$ and marginal distributions $g(x)$ and $h(y)$, respectively. The random variables X and Y are said to be **statistically independent** if and only if

$$f(x, y) = g(x)h(y)$$

for all (x, y) within their range.



The continuous random variables of Example 3.20 are statistically independent, since the product of the two marginal distributions gives the joint density function.

$$f(x, y) = \begin{cases} \frac{x(1+3y^2)}{4}, & 0 < x < 2, 0 < y < 1, \\ 0, & \text{elsewhere,} \end{cases}$$

$$g(x) = \frac{x}{2}, \quad h(y) = \frac{1+3y^2}{2}.$$

This is obviously not the case, however, for the continuous variables of Example 3.19.

$$f(x, y) = \begin{cases} 10xy^2, & 0 < x < y < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

$$g(x) = \frac{10}{3}x(1-x^3), \quad 0 < x < 1,$$

$$h(y) := 5y^4, \quad 0 < y < 1.$$



Checking for statistical independence of discrete random variables requires a more thorough investigation,

since it is possible to have the product of the marginal distributions equal to the joint probability distribution for some but not all combinations of (x, y) .

If you can find any point (x, y) for which $f(x, y)$ is defined such that $f(x, y) \neq g(x)h(y)$, the discrete variables X and Y are not statistically independent.



Example 3.21:

Show that the random variables of Example 3.14 are not statistically independent.

Proof: Let us consider the point $(0, 1)$.

From Table 3.1 we find the three probabilities

$f(0, 1)$, $g(0)$, and $h(1)$ to be

$$f(0, 1) = \frac{3}{14},$$

$$g(0) = \sum_{y=0}^2 f(0, y) = \frac{3}{28} + \frac{3}{14} + \frac{1}{28} = \frac{5}{14},$$

$$h(1) = \sum_{x=0}^2 f(x, 1) = \frac{3}{14} + \frac{3}{14} + 0 = \frac{3}{7}.$$

Clearly,

$$f(0, 1) \neq g(0)h(1),$$

and therefore X and Y are not statistically independent.

		x			Row Totals
		0	1	2	
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$	$\frac{15}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0	$\frac{3}{7}$
	2	$\frac{1}{28}$	0	0	$\frac{1}{28}$
Column Totals		$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$	1



All the preceding definitions concerning two random variables can be generalized to the case of n random variables.

Let $f(x_1, x_2, \dots, x_n)$ be the joint probability function of the random variables $X_1, X_2, \dots, X_n$.

The marginal distribution of X_1 , for example, is

$$g(x_1) = \sum_{x_2} \cdots \sum_{x_n} f(x_1, x_2, \dots, x_n)$$

for the discrete case, and

$$g(x_1) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(x_1, x_2, \dots, x_n) dx_2 dx_3 \cdots dx_n$$

for the continuous case.

We can now obtain **joint marginal distributions** such as $g(x_1, x_2)$, where

$$g(x_1, x_2) = \begin{cases} \sum_{x_3} \cdots \sum_{x_n} f(x_1, x_2, \dots, x_n) & \text{(discrete case),} \\ \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(x_1, x_2, \dots, x_n) dx_3 dx_4 \cdots dx_n & \text{(continuous case).} \end{cases}$$



We could consider numerous conditional distributions.

For example, the **joint conditional distribution** of X_1, X_2 , and X_3 , given that $X_4 = x_4, X_5 = x_5, \dots, X_n = x_n$, is written

$$f(x_1, x_2, x_3 \mid x_4, x_5, \dots, x_n) = \frac{f(x_1, x_2, \dots, x_n)}{g(x_4, x_5, \dots, x_n)},$$

where $g(x_4, x_5, \dots, x_n)$ is the joint marginal distribution of the random variables $X_4, X_5, \dots, X_n$.

A generalization of Definition 3.12 leads to the following definition for the mutual statistical independence of the variables $X_1, X_2, \dots, X_n$.



Definition 3.13:

Let $X_1, X_2, \dots, X_n$ be n random variables, discrete or continuous, with joint probability distribution $f(x_1, x_2, \dots, x_n)$ and marginal distribution $f_1(x_1), f_2(x_2), \dots, f_n(x_n)$, respectively. The random variables $X_1, X_2, \dots, X_n$ are said to be mutually **statistically independent** if and only if

$$f(x_1, x_2, \dots, x_n) = f_1(x_1)f_2(x_2) \cdots f_n(x_n)$$

for all $(x_1, x_2, \dots, x_n)$ within their range.



Example 3.22:

Suppose that the shelf life, in years, of a certain perishable food product packaged in cardboard containers is a random variable whose probability density function is given by

$$f(x) = \begin{cases} e^{-x}, & x > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

Let X_1 , X_2 , and X_3 represent the shelf lives for three of these containers selected independently

find $P(X_1 < 2, 1 < X_2 < 3, X_3 > 2)$.

Solution:

Since the containers were selected independently, we can assume that the random variables X_1 , X_2 , and X_3 are statistically independent, having the joint probability density

$$f(x_1, x_2, x_3) = f(x_1)f(x_2)f(x_3) = e^{-x_1}e^{-x_2}e^{-x_3} = e^{-x_1-x_2-x_3},$$

for $x_1 > 0$, $x_2 > 0$, $x_3 > 0$, and $f(x_1, x_2, x_3) = 0$ elsewhere.

Hence

$$\begin{aligned} P(X_1 < 2, 1 < X_2 < 3, X_3 > 2) &= \int_2^\infty \int_1^3 \int_0^2 e^{-x_1-x_2-x_3} dx_1 dx_2 dx_3 \\ &= (1 - e^{-2})(e^{-1} - e^{-3})e^{-2} = 0.0372. \end{aligned}$$

